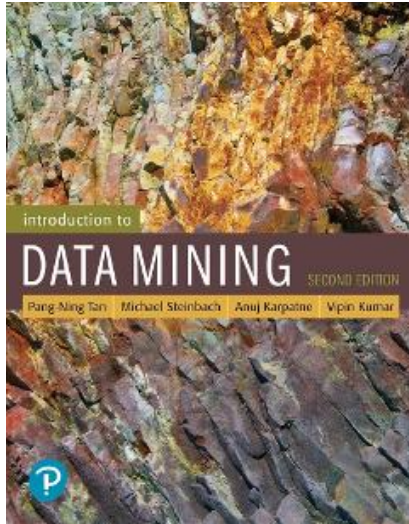


CSCE 5380/4380 – Data Mining



Dr. Moawia Eldow

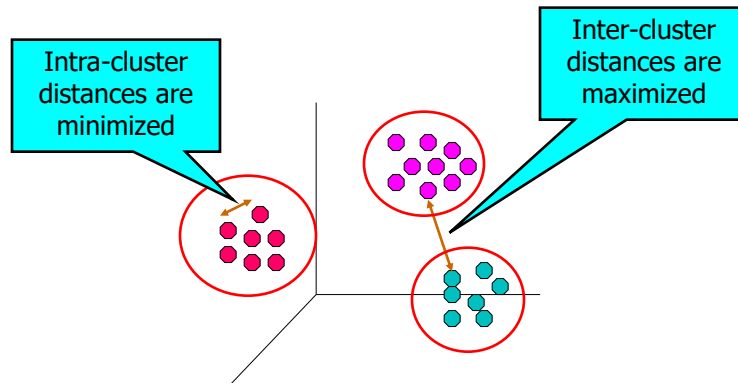
Chapter Seven:
**Cluster
Analysis: Basic
Concepts
and Algorithms**

Outline

- **Clustering Analysis**
 - Types of Clusterings
 - Types of Clusters
- **Clustering Algorithms**
 - Partitioning Clustering (K-means & variants)
 - Hierarchical Clustering (Agglomerative)
 - Density Based Clustering (DBSCAN)
- **Cluster Evaluation**

What is Cluster Analysis?

- Given a set of objects, place them in groups such that the objects in a group are similar (or related) to one another and different from (or unrelated to) the objects in other groups



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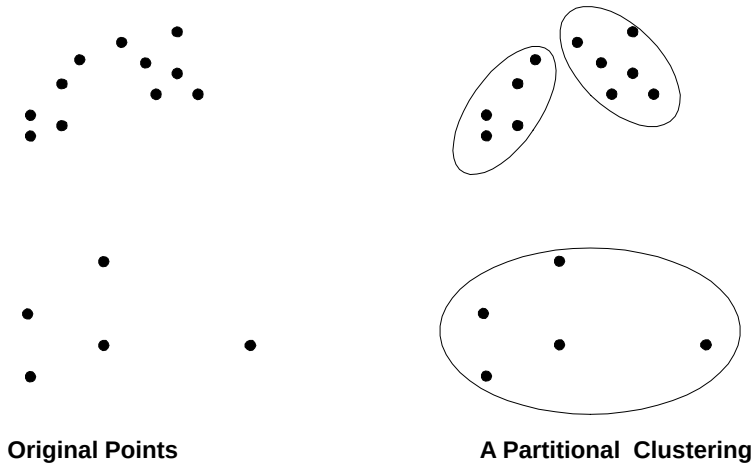
Types of Clusterings

- A **clustering** is a set of clusters
- Important distinction between **hierarchical** and **partitional** sets of clusters
 - Partitional Clustering
 - ◆ A division of data objects into non-overlapping subsets (clusters)
 - Hierarchical clustering
 - ◆ A set of nested clusters organized as a hierarchical tree

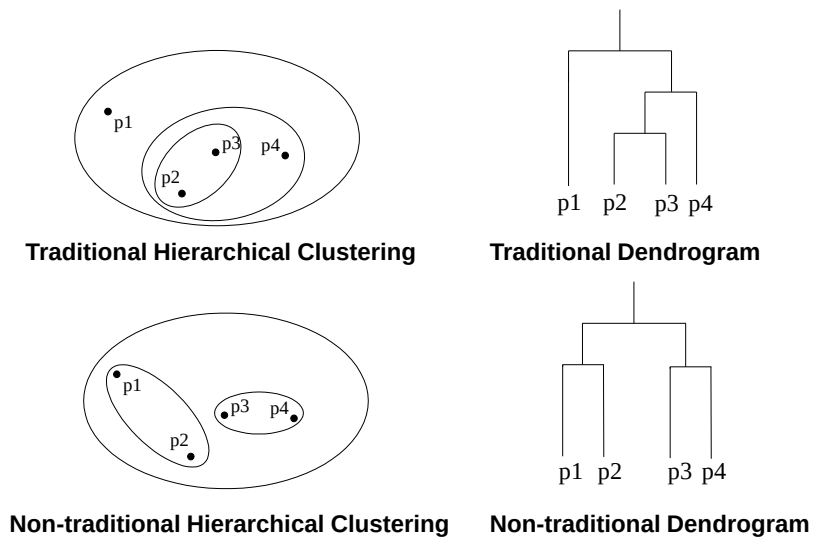
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Partitional Clustering



Hierarchical Clustering

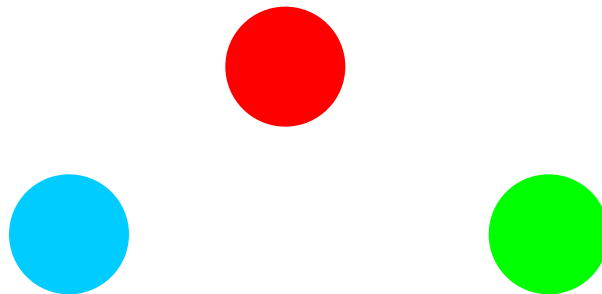


Types of Clusters

- Well-separated clusters
- Prototype-based clusters
- Contiguity-based clusters
- Density-based clusters
- Described by an Objective Function

Types of Clusters: Well-Separated

- Well-Separated Clusters:
 - A cluster is a set of points such that any point in a cluster is closer (or more similar) to every other point in the cluster than to any point not in the cluster.



3 well-separated clusters

Types of Clusters: Prototype-Based

□ Prototype-based

- A cluster is a set of objects such that an object in a cluster is closer (more similar) to the prototype or “center” of a cluster, than to the center of any other cluster
- The center of a cluster is often a **centroid**, the average of all the points in the cluster, or a **medoid**, the most “representative” point of a cluster



4 center-based clusters

Types of Clusters: Contiguity-Based

□ Contiguous Cluster (Nearest neighbor or Transitive)

- A cluster is a set of points such that a point in a cluster is closer (or more similar) to one or more other points in the cluster than to any point not in the cluster.

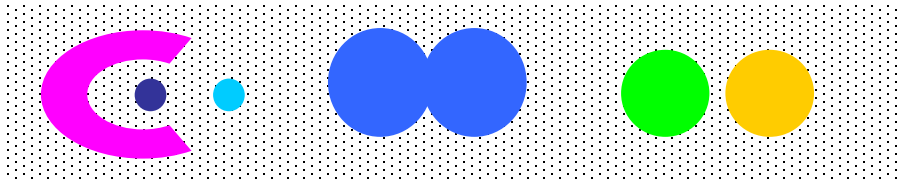


8 contiguous clusters

Types of Clusters: Density-Based

□ Density-based

- A cluster is a dense region of points, which is separated by low-density regions, from other regions of high density.
- Used when the clusters are irregular or intertwined, and when noise and outliers are present.



6 density-based clusters

Types of Clusters: Objective Function

□ Clusters Defined by an Objective Function

- Finds clusters that minimize or maximize an objective function.
- Enumerate all possible ways of dividing the points into clusters and evaluate the 'goodness' of each potential set of clusters by using the given objective function. (NP Hard)
- Can have global or local objectives.
 - ◆ Hierarchical clustering algorithms typically have local objectives
 - ◆ Partitional algorithms typically have global objectives
- A variation of the global objective function approach is to fit the data to a parameterized model.
 - ◆ Parameters for the model are determined from the data.
 - ◆ Mixture models assume that the data is a 'mixture' of a number of statistical distributions.

Characteristics of the Input Data Are Important

- Type of proximity or density measure
 - Central to clustering
 - Depends on data and application
- Data characteristics that affect proximity and/or density are
 - Dimensionality
 - ◆ Sparseness
 - Attribute type
 - Special relationships in the data
 - ◆ For example, autocorrelation
 - Distribution of the data
- Noise and Outliers
 - Often interfere with the operation of the clustering algorithm
- Clusters of differing sizes, densities, and shapes

Clustering Algorithms

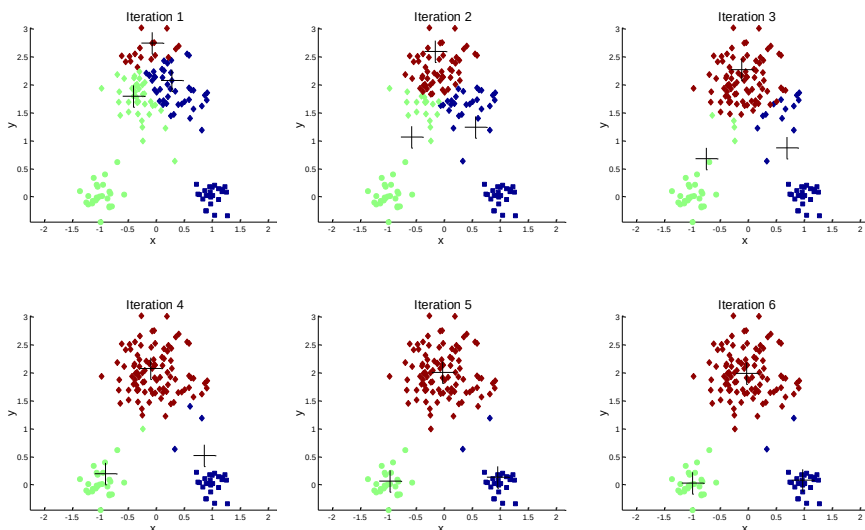
- K-means and its variants
(Covering of the algorithm)
- Hierarchical clustering
(covering of the algorithm)
- Density-based clustering
(Introducing the algorithm)

K-means Clustering

- Partitional clustering approach
- Number of clusters, K , must be specified
- Each cluster is associated with a **centroid** (center point)
- Each point is assigned to the cluster with the closest centroid
- The basic algorithm is very simple

-
- 1: Select K points as the initial centroids.
 - 2: **repeat**
 - 3: Form K clusters by assigning all points to the closest centroid.
 - 4: Recompute the centroid of each cluster.
 - 5: **until** The centroids don't change
-

Example of K-means Clustering



K-means Clustering – Details

- Simple iterative algorithm.
 - Choose initial centroids;
 - repeat {assign each point to a nearest centroid; re-compute cluster centroids}
 - until centroids stop changing.
- Initial centroids are often chosen randomly.
 - Clusters produced can vary from one run to another
- The centroid is (typically) the mean of the points in the cluster, but other definitions are possible (see Table 7.2).
- K-means will converge for common proximity measures with appropriately defined centroid (see Table 7.2)
- Most of the convergence happens in the first few iterations.
 - Often the stopping condition is changed to 'Until relatively few points change clusters'
- Complexity is $O(n * K * I * d)$
 - n = number of points, K = number of clusters,
 I = number of iterations, d = number of attributes

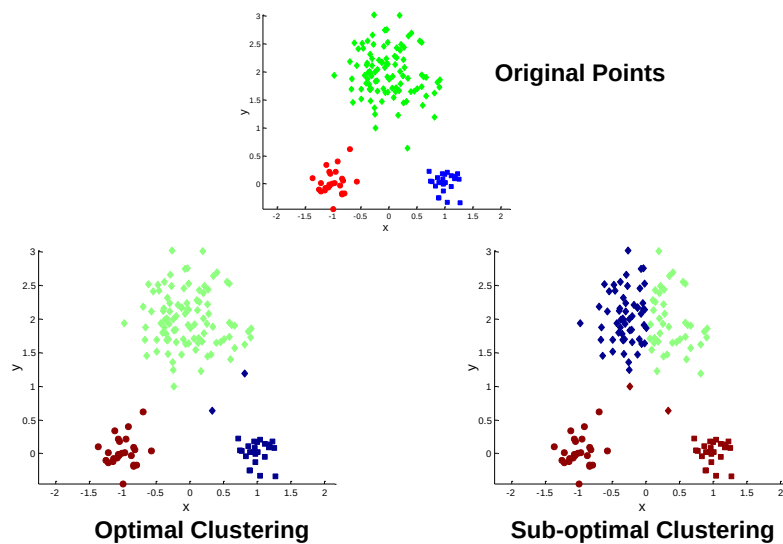
K-means Objective Function

- A common objective function (used with Euclidean distance measure) is Sum of Squared Error (SSE)
 - For each point, the error is the distance to the nearest cluster center
 - To get SSE, we square these errors and sum them.

$$SSE = \sum_{i=1}^K \sum_{x \in C_i} dist^2(m_i, x)$$

- x is a data point in cluster C_i and m_i is the centroid (mean) for cluster C_i
- SSE improves in each iteration of K-means until it reaches a local or global minima.

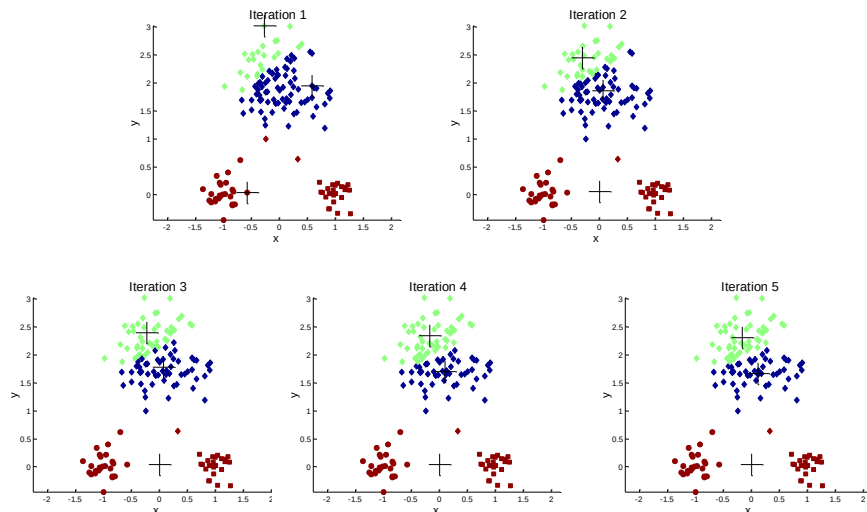
Two different K-means Clusterings



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Importance of Choosing Initial Centroids ...



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Problems with Selecting Initial Points

- If there are K 'real' clusters then the chance of selecting one centroid from each cluster is small.
 - Chance is relatively small when K is large
 - If clusters are the same size, n , then

$$P = \frac{\text{number of ways to select one centroid from each cluster}}{\text{number of ways to select } K \text{ centroids}} = \frac{K!n^K}{(Kn)^K} = \frac{K!}{K^K}$$

- For example, if $K = 10$, then probability = $10!/10^{10} = 0.00036$
- Sometimes the initial centroids will readjust themselves in 'right' way, and sometimes they don't

Solutions to Initial Centroids Problem

- Multiple runs
 - Helps, but probability is not on your side
- Use some strategy to select the k initial centroids and then select among these initial centroids
 - Select most widely separated
 - ◆ K-means++ is a robust way of doing this selection
 - Use hierarchical clustering to determine initial centroids
- Bisecting K-means
 - Not as susceptible to initialization issues

K-means++

- This approach can be slower than random initialization, but very consistently produces better results in terms of SSE
 - The k-means++ algorithm guarantees an approximation ratio $O(\log k)$ in expectation, where k is the number of centers
- To select a set of initial centroids, C , perform the following
 1. Select an initial point at random to be the first centroid
 2. For $k - 1$ steps
 3. For each of the N points, x_i , $1 \leq i \leq N$, find the minimum squared distance to the currently selected centroids, $C_1, \dots, C_j$, $1 \leq j < k$, i.e., $\min_j d^2(C_j, x_i)$
 4. Randomly select a new centroid by choosing a point with probability proportional to $\frac{\min_j d^2(C_j, x_i)}{\sum_i \min_j d^2(C_j, x_i)}$
 5. End For

Bisecting K-means

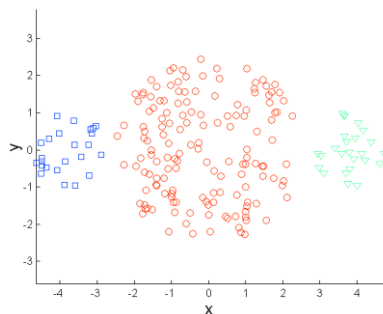
- Bisecting K-means algorithm
 - Variant of K-means that can produce a partitional or a hierarchical clustering

-
- 1: Initialize the list of clusters to contain the cluster containing all points.
 - 2: **repeat**
 - 3: Select a cluster from the list of clusters
 - 4: **for** $i = 1$ to *number_of_iterations* **do**
 - 5: Bisect the selected cluster using basic K-means
 - 6: **end for**
 - 7: Add the two clusters from the bisection with the lowest SSE to the list of clusters.
 - 8: **until** Until the list of clusters contains K clusters
-

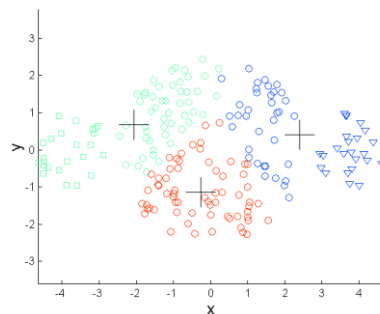
Limitations of K-means

- K-means has problems when clusters are of differing
 - Sizes
 - Densities
 - Non-globular shapes
- K-means has problems when the data contains outliers.
 - One possible solution is to remove outliers before clustering

Limitations of K-means: Differing Sizes

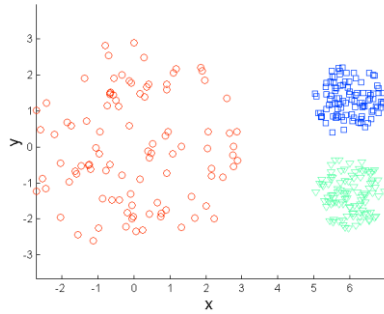


Original Points

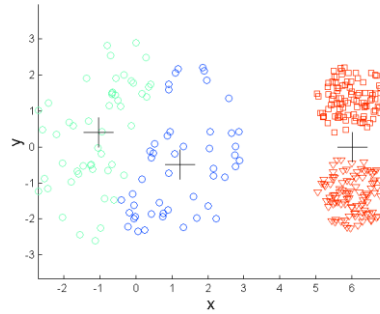


K-means (3 Clusters)

Limitations of K-means: Differing Density

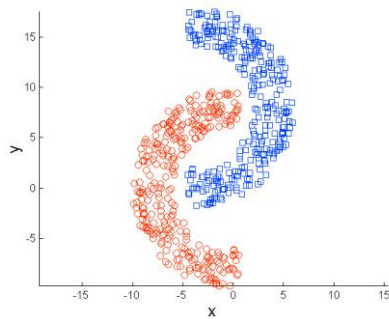


Original Points

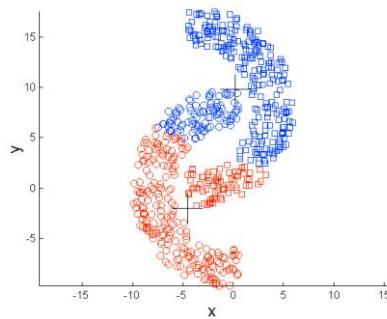


K-means (3 Clusters)

Limitations of K-means: Non-globular Shapes

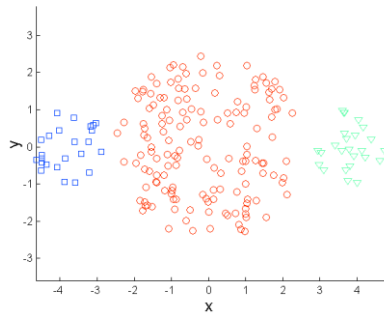


Original Points

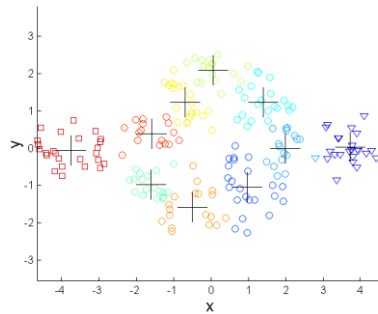


K-means (2 Clusters)

Overcoming K-means Limitations - Sizes



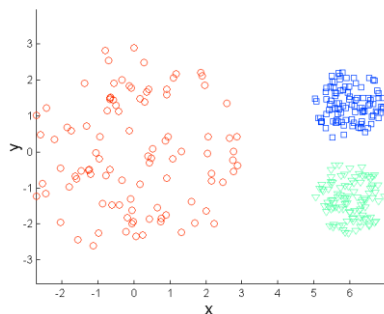
Original Points



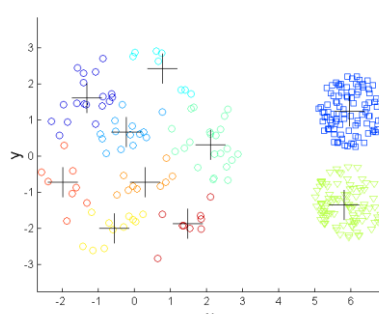
K-means Clusters

One solution is to find a large number of clusters such that each of them represents a part of a natural cluster. But these small clusters need to be put together in a post-processing step.

Overcoming K-means Limitations - Density



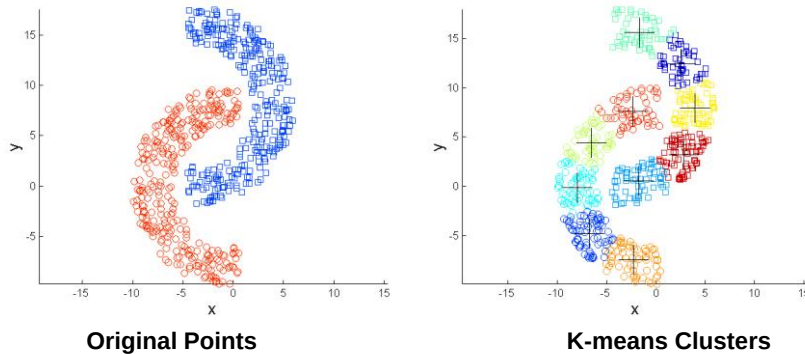
Original Points



K-means Clusters

One solution is to find a large number of clusters such that each of them represents a part of a natural cluster. But these small clusters need to be put together in a post-processing step.

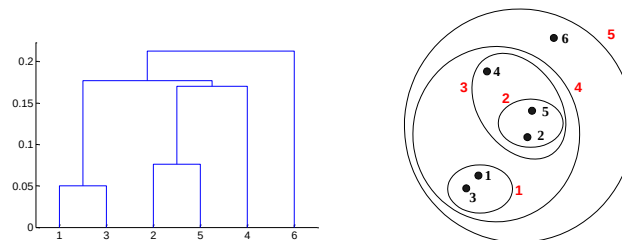
Overcoming K-means Limitations - Shapes



One solution is to find a large number of clusters such that each of them represents a part of a natural cluster. But these small clusters need to be put together in a post-processing step.

Hierarchical Clustering

- Produces a set of nested clusters organized as a hierarchical tree
- Can be visualized as a dendrogram
 - A tree like diagram that records the sequences of merges or splits



Strengths of Hierarchical Clustering

- Do not have to assume any particular number of clusters
 - Any desired number of clusters can be obtained by 'cutting' the dendrogram at the proper level
- They may correspond to meaningful taxonomies
 - Example in biological sciences (e.g., animal kingdom, phylogeny reconstruction, ...)

Hierarchical Clustering

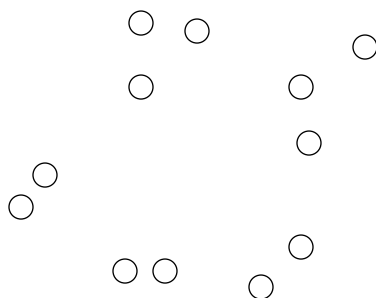
- Two main types of hierarchical clustering
 - Agglomerative:
 - ◆ Start with the points as individual clusters
 - ◆ At each step, merge the closest pair of clusters until only one cluster (or k clusters) left
 - Divisive:
 - ◆ Start with one, all-inclusive cluster
 - ◆ At each step, split a cluster until each cluster contains an individual point (or there are k clusters)
- Traditional hierarchical algorithms use a similarity or distance matrix
 - Merge or split one cluster at a time

Agglomerative Clustering Algorithm

- **Key Idea: Successively merge closest clusters**
- Basic algorithm
 1. Compute the proximity matrix
 2. Let each data point be a cluster
 3. **Repeat**
 4. Merge the two closest clusters
 5. Update the proximity matrix
 6. **Until** only a single cluster remains
- Key operation is the computation of the proximity of two clusters
 - Different approaches to defining the distance between clusters distinguish the different algorithms

Steps 1 and 2

- Start with clusters of individual points and a proximity matrix



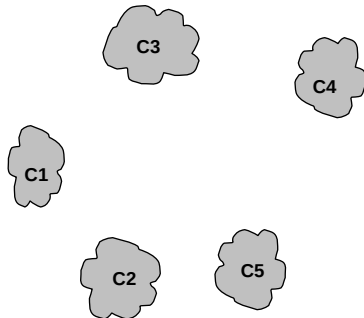
	p1	p2	p3	p4	p5	...
p1						
p2						
p3						
p4						
p5						
.						
.						
.						

Proximity Matrix



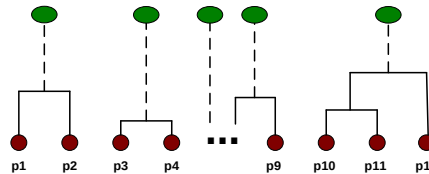
Intermediate Situation

- After some merging steps, we have some clusters



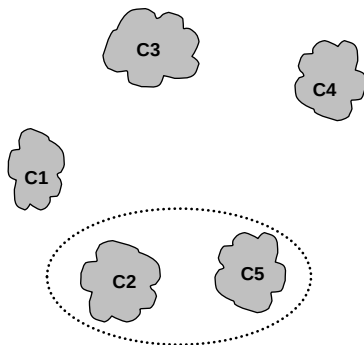
	C1	C2	C3	C4	C5
C1					
C2					
C3					
C4					
C5					

Proximity Matrix



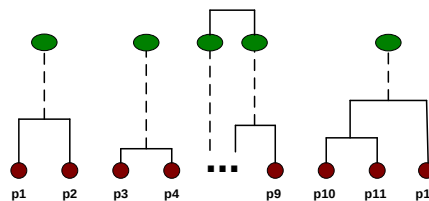
Step 4

- We want to merge the two closest clusters (C2 and C5) and update the proximity matrix.



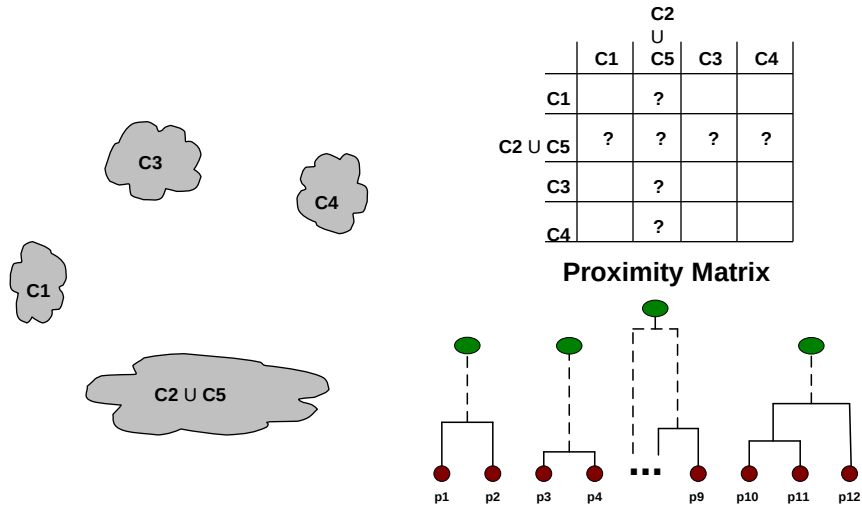
	C1	C2	C3	C4	C5
C1					
C2					
C3					
C4					
C5					

Proximity Matrix



Step 5

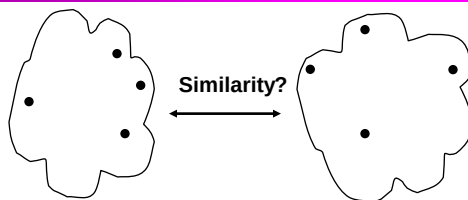
- The question is “How do we update the proximity matrix?”



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How to Define Inter-Cluster Distance



- MIN
- MAX
- Group Average
- Distance Between Centroids
- Other methods driven by an objective function
 - Ward's Method uses squared error

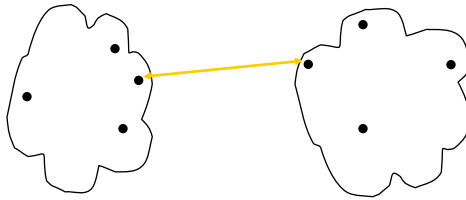
	p1	p2	p3	p4	p5	...
p1						
p2						
p3						
p4						
p5						
.						
.						
.						

Proximity Matrix

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How to Define Inter-Cluster Similarity

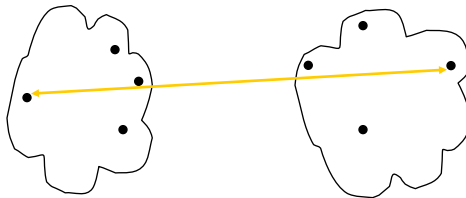


- MIN
- MAX
- Group Average
- Distance Between Centroids
- Other methods driven by an objective function
 - Ward's Method uses squared error

	p1	p2	p3	p4	p5	...
p1						
p2						
p3						
p4						
p5						
.						

Proximity Matrix

How to Define Inter-Cluster Similarity

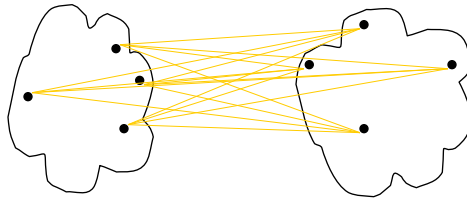


- MIN
- MAX
- Group Average
- Distance Between Centroids
- Other methods driven by an objective function
 - Ward's Method uses squared error

	p1	p2	p3	p4	p5	...
p1						
p2						
p3						
p4						
p5						
.						

Proximity Matrix

How to Define Inter-Cluster Similarity

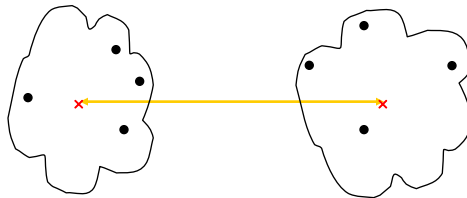


- MIN
- MAX
- **Group Average**
- Distance Between Centroids
- Other methods driven by an objective function
 - Ward's Method uses squared error

	p1	p2	p3	p4	p5	...
p1						
p2						
p3						
p4						
p5						
.						
.						
.						

Proximity Matrix

How to Define Inter-Cluster Similarity



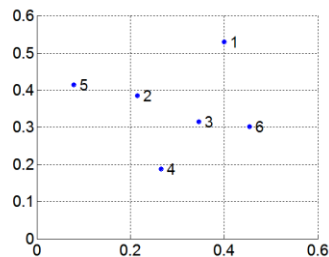
- MIN
- MAX
- Group Average
- **Distance Between Centroids**
- Other methods driven by an objective function
 - Ward's Method uses squared error

	p1	p2	p3	p4	p5	...
p1						
p2						
p3						
p4						
p5						
.						
.						
.						

Proximity Matrix

MIN or Single Link

- Proximity of two clusters is based on the two closest points in the different clusters
 - Determined by one pair of points, i.e., by one link in the proximity graph
- Example:



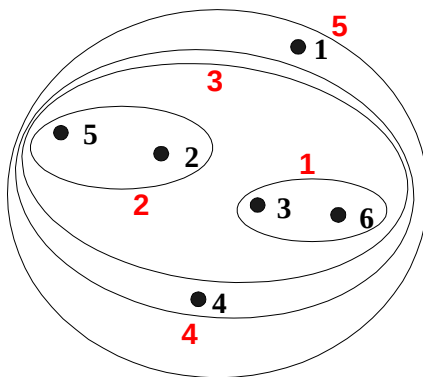
Distance Matrix:

	p1	p2	p3	p4	p5	p6
p1	0.00	0.24	0.22	0.37	0.34	0.23
p2	0.24	0.00	0.15	0.20	0.14	0.25
p3	0.22	0.15	0.00	0.15	0.28	0.11
p4	0.37	0.20	0.15	0.00	0.29	0.22
p5	0.34	0.14	0.28	0.29	0.00	0.39
p6	0.23	0.25	0.11	0.22	0.39	0.00

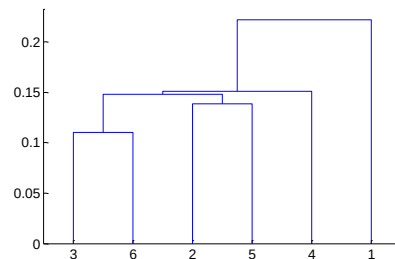
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Hierarchical Clustering: MIN



Nested Clusters

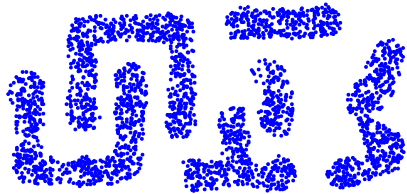


Dendrogram

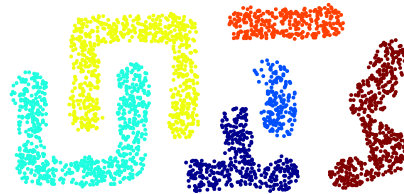
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Strength of MIN



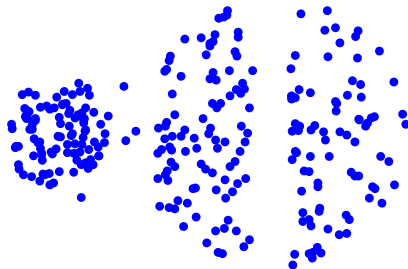
Original Points



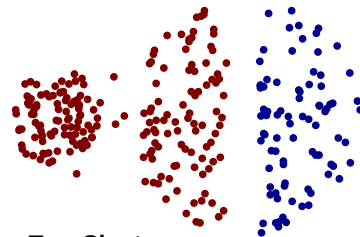
Six Clusters

- Can handle non-elliptical shapes

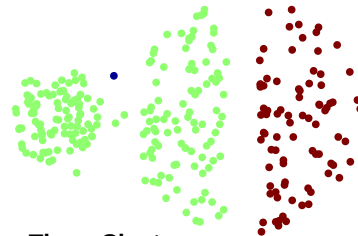
Limitations of MIN



Original Points



Two Clusters

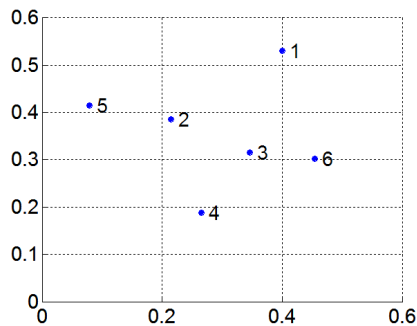


Three Clusters

- Sensitive to noise

MAX or Complete Linkage

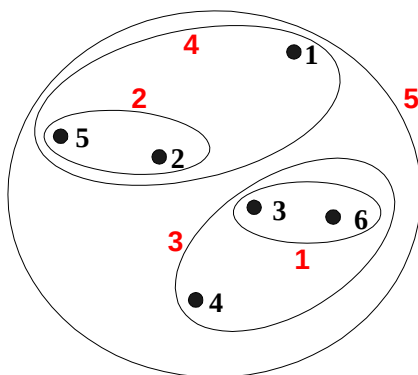
- Proximity of two clusters is based on the two most distant points in the different clusters
 - Determined by all pairs of points in the two clusters



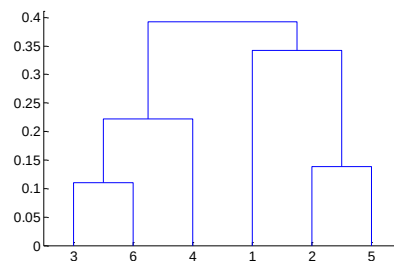
Distance Matrix:

	p1	p2	p3	p4	p5	p6
p1	0.00	0.24	0.22	0.37	0.34	0.23
p2	0.24	0.00	0.15	0.20	0.14	0.25
p3	0.22	0.15	0.00	0.15	0.28	0.11
p4	0.37	0.20	0.15	0.00	0.29	0.22
p5	0.34	0.14	0.28	0.29	0.00	0.39
p6	0.23	0.25	0.11	0.22	0.39	0.00

Hierarchical Clustering: MAX

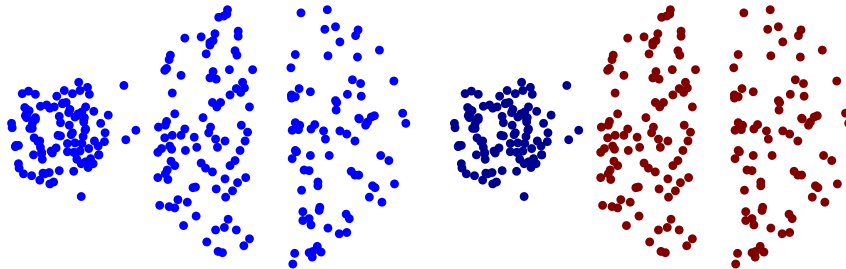


Nested Clusters



Dendrogram

Strength of MAX

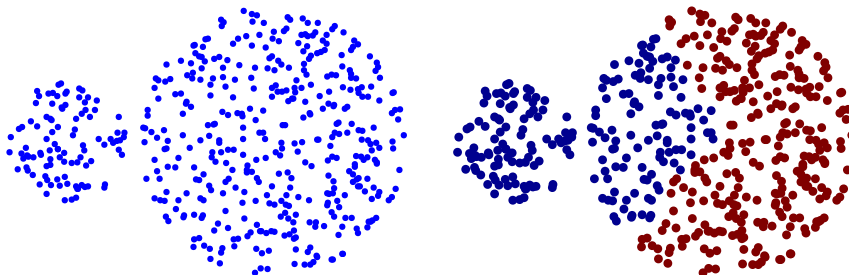


Original Points

Two Clusters

- Less susceptible to noise

Limitations of MAX



Original Points

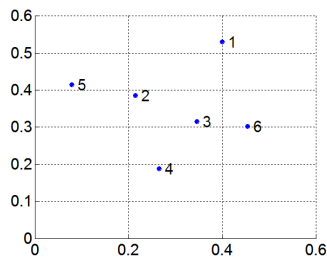
Two Clusters

- Tends to break large clusters
- Biased towards globular clusters

Group Average

- Proximity of two clusters is the average of pairwise proximity between points in the two clusters.

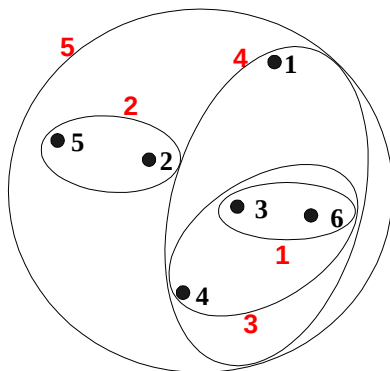
$$\text{proximity}(\text{Cluster}_i, \text{Cluster}_j) = \frac{\sum_{\substack{p_i \in \text{Cluster}_i \\ p_j \in \text{Cluster}_j}} \text{proximity}(p_i, p_j)}{|\text{Cluster}_i| \times |\text{Cluster}_j|}$$



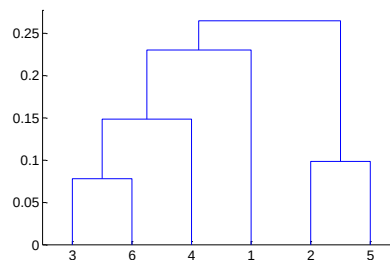
Distance Matrix:

	p1	p2	p3	p4	p5	p6
p1	0.00	0.24	0.22	0.37	0.34	0.23
p2	0.24	0.00	0.15	0.20	0.14	0.25
p3	0.22	0.15	0.00	0.15	0.28	0.11
p4	0.37	0.20	0.15	0.00	0.29	0.22
p5	0.34	0.14	0.28	0.29	0.00	0.39
p6	0.23	0.25	0.11	0.22	0.39	0.00

Hierarchical Clustering: Group Average



Nested Clusters



Dendrogram

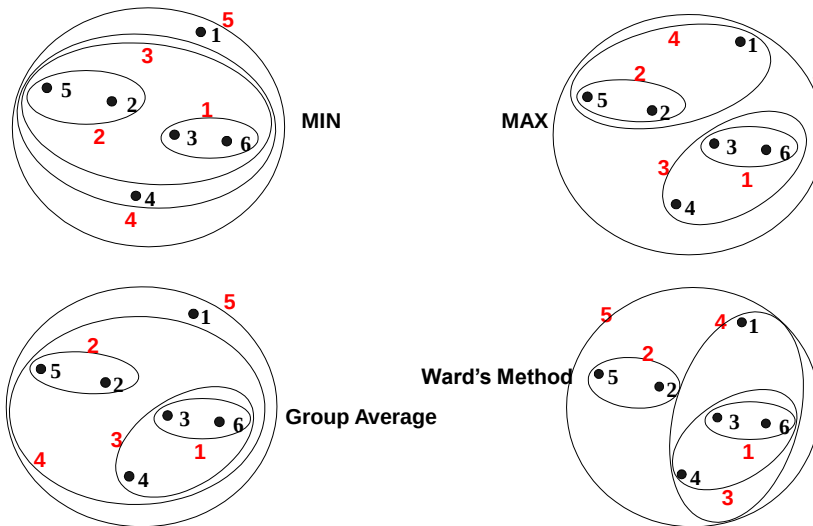
Hierarchical Clustering: Group Average

- Compromise between Single and Complete Link
- Strengths
 - Less susceptible to noise
- Limitations
 - Biased towards globular clusters

Cluster Similarity: Ward's Method

- Similarity of two clusters is based on the increase in squared error when two clusters are merged
 - Similar to group average if distance between points is distance squared
- Less susceptible to noise
- Biased towards globular clusters
- Hierarchical analogue of K-means
 - Can be used to initialize K-means

Hierarchical Clustering: Comparison



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Hierarchical Clustering: Problems and Limitations

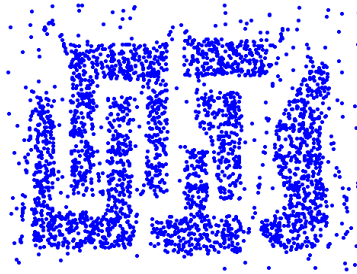
- Once a decision is made to combine two clusters, it cannot be undone
- No global objective function is directly minimized
- Different schemes have problems with one or more of the following:
 - Sensitivity to noise
 - Difficulty handling clusters of different sizes and non-globular shapes
 - Breaking large clusters

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Density Based Clustering

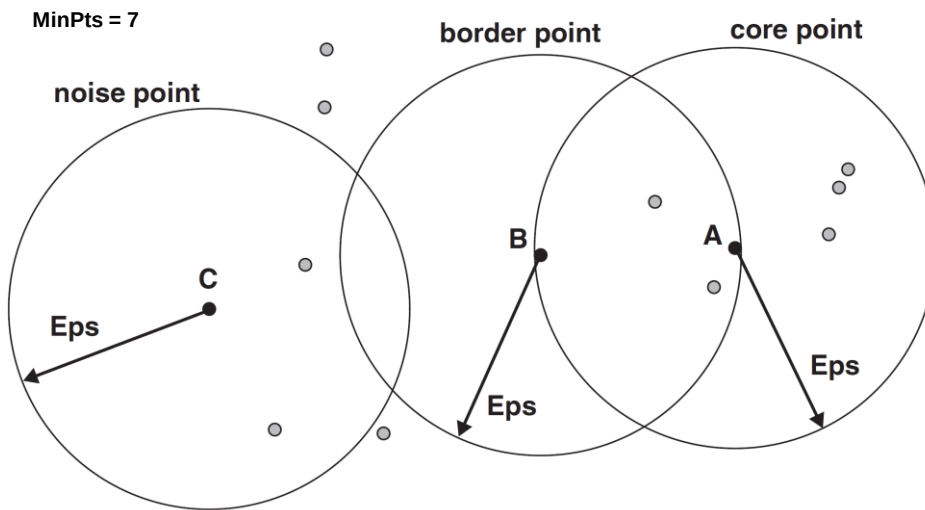
- Clusters are regions of high density that are separated from one another by regions of low density.



DBSCAN

- DBSCAN is a density-based algorithm.
 - Density = number of points within a specified radius (Eps)
 - A point is a **core point** if it has at least a specified number of points (MinPts) within Eps
 - ◆ These are points that are at the interior of a cluster
 - ◆ Counts the point itself
 - A **border point** is not a core point, but is in the neighborhood of a core point
 - A **noise point** is any point that is not a core point or a border point

DBSCAN: Core, Border, and Noise Points



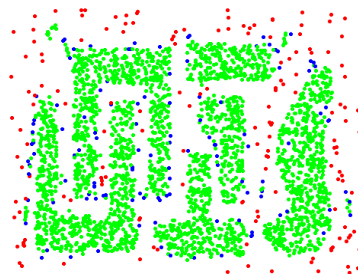
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DBSCAN: Core, Border and Noise Points



Original Points



Point types: **core**,
border and **noise**

Eps = 10, MinPts = 4

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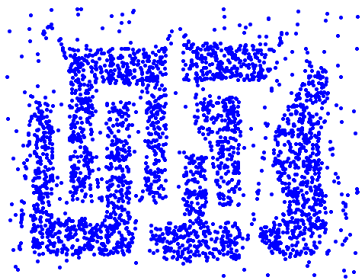
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DBSCAN Algorithm

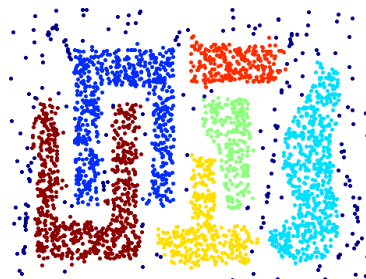
- Form clusters using core points, and assign border points to one of its neighboring clusters

- 1: Label all points as core, border, or noise points.
- 2: Eliminate noise points.
- 3: Put an edge between all core points within a distance Eps of each other.
- 4: Make each group of connected core points into a separate cluster.
- 5: Assign each border point to one of the clusters of its associated core points

When DBSCAN Works Well



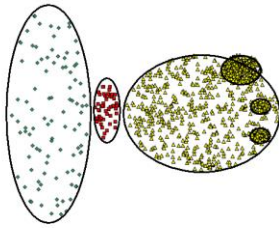
Original Points



Clusters (dark blue points indicate noise)

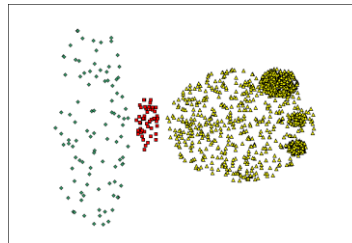
- Can handle clusters of different shapes and sizes
- Resistant to noise

When DBSCAN Does NOT Work Well

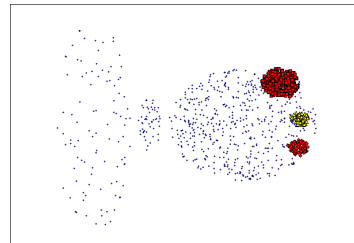


Original Points

- Varying densities
- High-dimensional data



(MinPts=4, Eps=9.92).

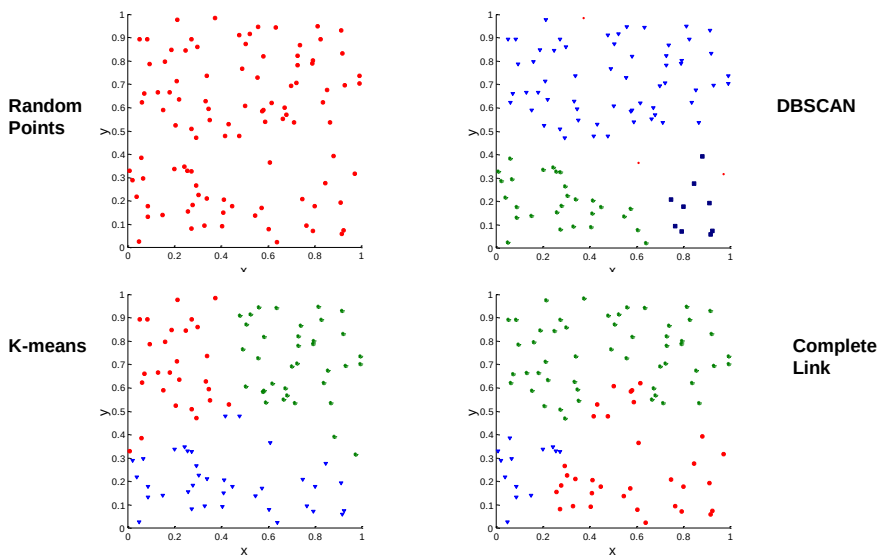


(MinPts=4, Eps=9.75)

Cluster Evaluation (Cluster Validity)

- For supervised classification we have a variety of measures to evaluate how good our model is
 - Accuracy, precision, recall
- For cluster analysis, the analogous question is how to evaluate the “goodness” of the resulting clusters?
- But “clusters are in the eye of the beholder”!
 - In practice the clusters we find are defined by the clustering algorithm
- Then why do we want to evaluate them?
 - To avoid finding patterns in noise
 - To compare clustering algorithms
 - To compare two sets of clusters
 - To compare two clusters

Clusters found in Random Data



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Measures of Cluster Validity

- Numerical measures that are applied to judge various aspects of cluster validity, are classified into the following two types.
 - **Supervised:** Used to measure the extent to which cluster labels match externally supplied class labels.
 - ◆ Entropy
 - ◆ Often called *external indices* because they use information external to the data
 - **Unsupervised:** Used to measure the goodness of a clustering structure *without* respect to external information.
 - ◆ Sum of Squared Error (SSE)
 - ◆ Often called *internal indices* because they only use information in the data
- You can use supervised or unsupervised measures to compare clusters or clusterings

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Unsupervised Measures: Cohesion and Separation

- **Cluster Cohesion:** Measures how closely related are objects in a cluster
 - Example: SSE
- **Cluster Separation:** Measure how distinct or well-separated a cluster is from other clusters
- Example: Squared Error
 - Cohesion is measured by the within cluster sum of squares (SSE)

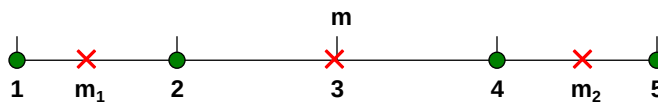
$$SSE = \sum_i \sum_{x \in C_i} (x - m_i)^2$$
 - Separation is measured by the between cluster sum of squares

$$SSB = \sum_i |C_i| (m - m_i)^2$$

Where $|C_i|$ is the size of cluster i

Unsupervised Measures: Cohesion and Separation

- Example: SSE
 - $SSB + SSE = \text{constant}$



K=1 cluster:

$$SSE = (1 - 3)^2 + (2 - 3)^2 + (4 - 3)^2 + (5 - 3)^2 = 10$$

$$SSB = 4 \times (3 - 3)^2 = 0$$

$$Total = 10 + 0 = 10$$

K=2 clusters:

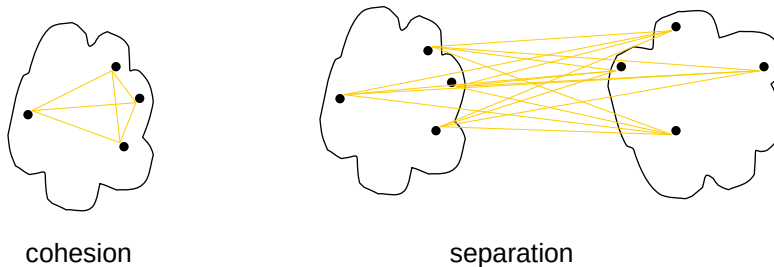
$$SSE = (1 - 1.5)^2 + (2 - 1.5)^2 + (4 - 4.5)^2 + (5 - 4.5)^2 = 1$$

$$SSB = 2 \times (3 - 1.5)^2 + 2 \times (4.5 - 3)^2 = 9$$

$$Total = 1 + 9 = 10$$

Unsupervised Measures: Cohesion and Separation

- A proximity graph-based approach can also be used for cohesion and separation.
 - Cluster cohesion is the sum of the weight of all links within a cluster.
 - Cluster separation is the sum of the weights between nodes in the cluster and nodes outside the cluster.

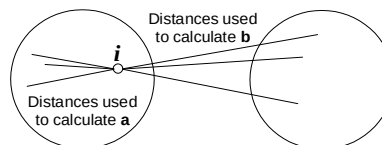


Unsupervised Measures: Silhouette Coefficient

- Silhouette coefficient combines ideas of both cohesion and separation, but for individual points, as well as clusters and clusterings
- For an individual point, i
 - Calculate a = average distance of i to the points in its cluster
 - Calculate b = min (average distance of i to points in another cluster)
 - The silhouette coefficient for a point is then given by

$$s = (b - a) / \max(a, b)$$

- Value can vary between -1 and 1
- Typically ranges between 0 and 1.
- The closer to 1 the better.



- Can calculate the average silhouette coefficient for a cluster or a clustering